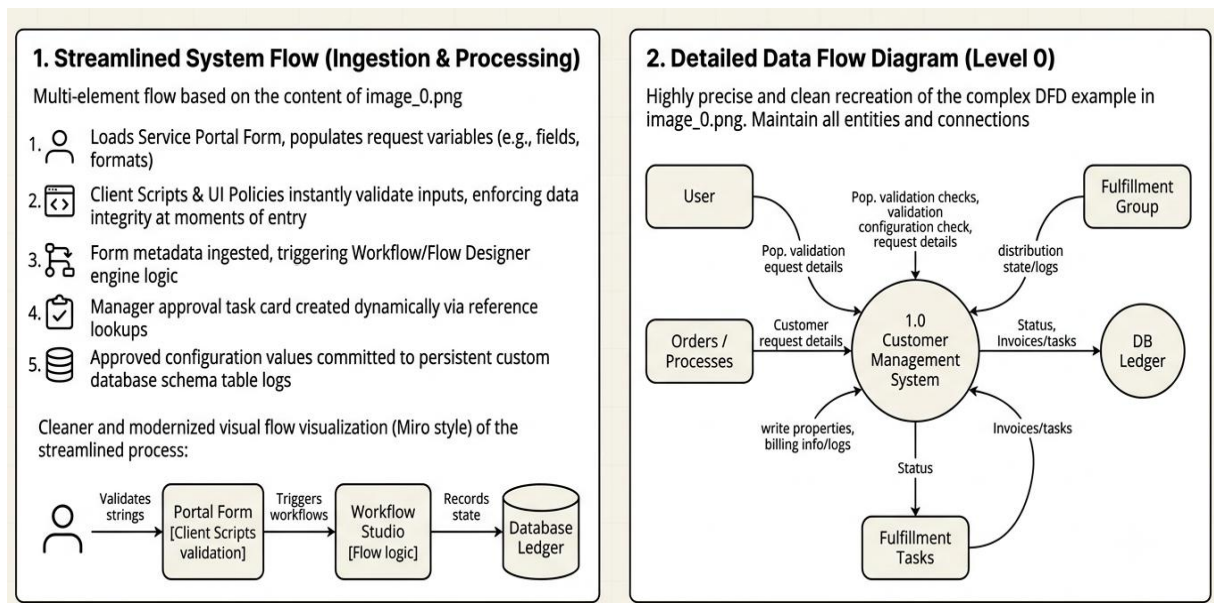


Project Design Phase-II

Data Flow Diagram & User Stories

Date	23 June 2026
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Project Name	Automated Network Request Management System
Maximum Marks	4 marks

Unified Data Flow Overview and DFD



2. Project User Stories Matrix

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Customer (Web user)	Request Ingestion Portal	USN-1	As a user, I can submit a network connectivity or device request through a central Service Portal catalog item so that I don't use manual emails or chat lines.	Given the user opens the catalog item, when they populate parameters, then fields must enforce strict alphanumeric validations before submission.	High	Sprint-1
Customer (Web user)	Session Automation Profile	USN-2	As a user, I want the ingestion form to	Given the form runtime context initializes, when the	High	Sprint-1

			automatically populate my name, contact email, and department using my active login session metadata.	system reads the active session ID, then it must auto-bind user profile variables without typing.		
Customer (Web user)	Progress Visibility	USN-3	As a user, I can track my active ticket progress stage via a real-time progress bar rendering on the portal dashboard header.	Given a submitted request ticket, when the backend process state updates, then the client-side portal view tracking line shifts instantly.	Medium	Sprint-1
Manager	Approval Routing Engine	USN-4	As a manager, I can receive a direct approval request card summary notification so that I can click to authorize or reject requests cleanly.	Given a validated request submission, when Flow Designer matches corporate hierarchies, then it must spawn interactive task cards with action options.	High	Sprint-1
Network Analyst	Fulfillment Workspace Management	USN-5	As an engineer, I can work out of an engineering task queue containing 100% complete validated parameters so that I eliminate back-and-forth loops.	Given an assigned card workspace, when an engineer opens the fulfillment view, then all required logical strings and port metrics must be locked and formatted.	High	Sprint-1
Network Analyst	Custom Schema Execution	USN-6	As an engineer, I can update the network status records out of specific child task layouts (u_network_task) linked to our structural data ledger table.	Given a task state shift to closed, when the updates fire, then system business rules must mirror and commit updates cleanly to u_network_database.	Medium	Sprint-2
Administrator	Compliance Ledger Tracking	USN-7	As an administrator, I can review	Every workflow transition state change is captured	Low	Sprint-2

			unalterable system history tracking streams of all ticket state variations, approvals, and execution records.	securely inside sys_audit tables.		
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Agile Sprint Estimation & Priority Summary

To ensure a structured implementation of the **Automated Network Request Management System** within the ServiceNow environment, the user stories mapped above are broken down into active execution sprint cycles based on platform dependencies:

- **Sprint 1 Focus (Core Infrastructure & Intake):** Focuses heavily on the user-facing catalog item interface (USN-1, USN-2), immediate status visualization features (USN-3), and the core automated approval routing mechanics (USN-4) running inside Flow Designer. This builds the fundamental path data must travel before engineers interact with it.
- **Sprint 2 Focus (Fulfillment & Compliance):** Moves into the backend engineering workspace layouts (USN-5, USN-6). This ensures closed child tasks cleanly commit request values directly to the persistent custom schema database ledger (u_network_database) while preserving unalterable audit trails (USN-7) inside the platform-native audit files (sys_audit).

Traceability Mapping: DFD to User Stories

Every functional user story directly maps to a data flow movement identified in the Level 0 Data Flow Diagram (DFD), ensuring complete architectural alignment:

- **Data Ingestion (DFD Process 1.0):** Powered directly by USN-1 and USN-2, managing the transition of raw user requirement strings from the browser interface into structured platform variables.
- **Validation Checks & Approvals (DFD Data Flows):** Governed by USN-4 and USN-5, where the Flow Designer engine acts as the primary gatekeeper, blocking data entry errors and routing tasks to corporate managers.
- **Data Ledger Storage (DFD Database Store):** Executed by USN-6 and tracked by USN-7, which handles the final, permanent write operations to custom relational tables, converting temporary requests into immutable historical assets.